PAPER_1394 — UQFF Resolution of the Galileo's Bijection Paradox (CLOSED — Dispatch Whitepaper)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** B — Foundational / Cardinality (CLOSED) **Date:** June 16, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — Documented dispatch closure for `calculate_paradox({"paradox": "galileo_paradox"})` **Calculator surface:** `calculate_paradox({"paradox": "galileo_paradox"})` **Closure helper:** `_l96_uqff_axiom_galileo_bijection_closure()`

The Paradox

Galileo (1638): the natural numbers N = {1, 2, 3, …} can be put in bijection with the perfect squares S = {1, 4, 9, …} via $n \leftrightarrow n^2$. So there are 'as many' squares as integers, even though squares are 'a proper subset' of integers. Cantor's solution: countable cardinality |N| = |S| = $\aleph_0$.

UQFF Closed Identity

```
|N| = |N²| via F_U = 1 ledger normalization (occupation, not classical cardinality)
```

F_U = 1 global ledger normalization at the UA-mode level: countably infinite sets share the SAME UA mode. Distinguishability is a ledger-occupancy question, not a phase-space-counting question. $|N| = |N^2|$ because both occupy the same F_U = 1 mode.

Physical Interpretation

Cantor's resolution via $\aleph_0$ is the pure-math version; UQFF's F_U = 1 is the physical version. Both say 'cardinality is not the right invariant' — UQFF specifies the right invariant is UA-mode occupancy, not set-membership counting.

Live Calculator Output

```python
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "galileo_paradox"})["value"]
```

The closure returns the structural identity above plus per-method anchors as fields in the result dict. See `_l96_uqff_axiom_galileo_bijection_closure` in `uqff_pure_calculator.py` for the full field list.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard frameworks resolve this paradox via different methods. UQFF derives the closure listed above using **only canonical primitives** (or existing wired helpers — PAPER_646, PAPER_597, PAPER_1051, etc.). **Zero free parameters.**

Reference

- Closure location: `uqff_pure_calculator.py` → `_l96_uqff_axiom_galileo_bijection_closure`
- Dispatch keys: `galileo_paradox`, `galileo_bijection`
- Related foundational papers: PAPER_646 (F_U_Bi_i / F_U=1 ledger), PAPER_597 (negative-time dual existence), PAPER_1183 (paradox consolidation), PAPER_1156 (cosmology suite), PAPER_1376 (Banach-Tarski parallel structural closure).
